

Solution Design

YAT Enrolline Cloud Architecture — Approved Improvement Design

Engagement	YAT Enrolline Cloud Infrastructure Improvement
Document type	Technical design (Solution Design)
Version	v1.0 — Approved for implementation
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Approved by	Sam Walker (YAT ICT Manager) · Jess Tran (YAT Registrar) · Asha Rao (YAT Compliance)
Supersedes	Nothing — it improves the Baseline Design, which remains the as-was record
Classification	Internal — YAT ICT, Student Administration, Compliance, and MTS personnel on signed MSA

Contents

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1. Purpose and Scope

This document specifies the improvements YAT approved at the Phase 1 design review, and is the design the Improvement Team implements in Phase 2. It takes the single-Availability-Zone Enrolline baseline and improves it across security, reliability, scalability and cost, plus a light India residency slice for regulatory logs.

The work is divided across four components — network, compute, database and storage — one per Improvement Team member. The Enrolline application and its student records are preserved unchanged throughout.

2. Approved Improvements by Component

2.1 Network

Change	Why
Add application subnet enrolline-app-b in ap-southeast-2b	Gives the compute tier somewhere to place second-zone capacity
Add a Gateway VPC endpoint for Amazon S3	Document traffic stops leaving the VPC through the NAT Gateway — removes a data-processing charge and a dependency
Enable VPC flow logs to CloudWatch Logs	Required to detect and investigate a personal-data breach; no network record exists today
Enable load-balancer access logs to S3	Access record for the staff-facing entry point

2.2 Compute

Change	Why
Extend the Auto Scaling group across enrolline-app-a and enrolline-app-b	Removes the single-zone application failure mode
Raise max capacity to 4	Headroom for the intake peak once capacity is no longer held constantly
Add scheduled scaling against the academic calendar	The peaks are published months ahead; holding peak capacity for forty-six quiet weeks is the largest identified inefficiency

Scheduled scaling raises desired capacity ahead of each intake enrolment window and each census date, and returns it afterwards. The schedule is derived from the published academic calendar, not from observed load, so capacity is in place before the load arrives.

2.3 Database

Change	Why
Convert the RDS instance to a Multi-AZ deployment with an automatic-failover standby	Brings recovery inside the two-hour objective, which point-in-time restore does not reliably meet
Raise automated backup retention to 14 days	Wider recovery window across a full intake cycle

Multi-AZ is a synchronous standby in the second Availability Zone with automatic failover. It is not a read replica and adds no read capacity; it is bought for recovery time, and its cost is approximately a doubling of the database line.

Building this in a lab the lab role may refuse `rds:ModifyDBInstance`. Where it does, the Multi-AZ conversion is specified and justified in the design and verified by inspection of the template rather than by a live change. Say which you did.

2.4 Storage

Change	Why
Add an S3 lifecycle configuration — Standard → Standard-IA at 90 days → Glacier Instant Retrieval at 2 years	Attachment reads fall away once an enrolment closes, but 30-year retention means nothing is ever deleted
Enable S3 server access logging	Access record over student identity evidence
Confirm Block Public Access and versioning remain enforced	No change intended; verified as part of the improvement

The lifecycle configuration does not delete anything. Retention is unchanged at 30 years; only the storage class changes as objects age.

3. India Residency Slice

The CERT-In Directions require system and access logs for the India-related operation to be held within Indian jurisdiction for a rolling 180 days and to be retrievable quickly enough to support a six-hour incident-reporting obligation.

- System and access logs for the India operation are replicated to a log destination in the Mumbai region (ap-south-1) with 180-day retention.
- The main system, the student records and the document store remain in Sydney — the DPDP Act permits this, and the Australian residency position is unchanged.
- Only logs cross the border. No student personal data is moved to India by this design.

Not applicable — no deployable lab variant is produced for the residency slice; it is specified at design level and assessed as a design decision.

4. Goals and Performance Metrics

Goal	Metric	Target
Reliability	Recovery time from database failure	≤ 2 hours, evidenced by failover
Reliability	Availability during an intake window	99.9%
Scalability	Capacity available at the start of an intake window	In place before the window opens, by schedule
Security	Network and access logging coverage	Flow logs, ALB access logs, S3 access logs all enabled
Cost	Compute cost outside intake windows	Reduced against the year-round peak-capacity baseline

5. What Is Deliberately Not Changed

- The Enrolline application and its PostgreSQL engine — IR-4.
- The Sydney region for all student records and documents.
- The commercial Enrolline licensing and vendor support, which are fixed against the user population and unaffected by infrastructure change.
- Retention periods for any record class.

6. Implementation and Change Control

All production-affecting work is applied as a change-set against the deployed baseline and scheduled outside the Restricted Period — the intake enrolment windows, the week either side of a census date, and the January AVETMISS submission window. Every change in this design is additive or in-place; none replaces the database or moves data.

7. References

- Enrolline Cloud Architecture — Baseline Design — the as-was state this design improves
- Improvement Requirements (IR-1...7) — the outcomes this design is justified against
- Indian Regulatory Requirements — the obligations behind §3
- Enrolline Operational Costing — the cost base each improvement is measured against
- Change Management Procedure (intranet policies)

Document control

Field	Value
Document owner	Pat Lin, MTS Senior Consultant
Business owner	Jess Tran, YAT Registrar
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Review cycle	On material change to the architecture or its operating profile
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